Appendix for: The impact of frailty on COVID-19 outcomes: a systematic review and Meta-analysis of 16 cohort studies

1. Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Guidelines – Checklist.

Criteria		Brief description of how the criteria were handled in the meta-analysis
√	Problem definition	Recent observational studies have shown that frail COVID-19 patients are at increased risk of death. Frailty is not well recognized in the COVID-19 literature. To understand the evidence in its entirety, there is a need for a systematic review and meta-analysis.
√	Hypothesis statement	We hypothesized that frailty would be associated with higher all-cause mortality, COVID-19 severity, admission to intensive care unit, application of invasive mechanical ventilation, long-length stay
√	Description of study outcomes	Hospital mortality, long-term mortality (>6 months), ICU mortality, health resource utilization (e.g. hospital and ICU length of stay, receipt of mechanical ventilation, receipt of vasoactive medication), and discharge disposition.
√	Type of exposure or intervention used	Frail and non-frail patients with COVID-19.
√	Type of study designs used	Cohort studies were considered for inclusion. Narrative reviews, case reports, case-series, and editorials were excluded.
√	Study population	Patients with COVID-19
Reporting of search strategy should include		
√	Qualifications of searchers	The credentials of the authors are indicated in the author list. The contributions of the authors is listed in the manuscript..
√	Search strategy, including time period included in the synthesis and keywords	Medline (inception – October 2020), Embase (inception – October 2020), Pubmed (inception – October 2020) databases and Chinese databases(China National knowledge Infrastructure,Wangfang and Weipu)(inception – October 2020)) . Please see the Search strategy in the Appendix. The terms: “Frail”, “Frailty”, “Frail elderly”, “SARS-COV-2” and “COVID-19” were used with Boolean operators “OR” as well as “AND” in various combinations.
√	Databases and registries searched	Medline, Embase, Pubmed, and Chinese databases(China National knowledge Infrastructure,Wangfang and Weipu). Please see the Search strategy in the Appendix.
√	Search software used, name and version, including special features	We did not employ search software. EndNote was used to merge retrieved citations and eliminate duplications.
√	Use of hand searching	We hand-searched bibliographies of retrieved papers for additional references.
√	List of citations located and those excluded, including justifications	Details of the literature search process are outlined in the Search strategy in the Appendix. A citation list is available upon request.
√	Method of addressing articles published in languages other than English	We placed no restrictions on language. However, all articles considered for inclusion were published in English and Chinese.
√	Method of handling abstracts and unpublished studies	Abstracts were included. Authors of unpublished studies and abstracts were contacted for clarifying information, or additional information when necessary. We contacted authors for full report of relevant unpublished studies.
√	Description of any contact with authors	Full text of all included studies could be obtain,so it is not necessary to contact the authors.
Reporting of methods should include		
√	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	Inclusion and exclusion criteria are described in the section of Methods.
√	Rationale for the selection and coding of data	Data extracted were bibliographic details of included studies, region, age, sample, study design, percentage of male, the source of patients, definitions of frailty, outcome in frail and non-frail populations.

√	Assessment of confounding	We reported adjusted results and pooled adjusted results where possible.
√	Assessment of study quality, including blinding of quality assessors; stratification or regression on possible predictors of study results	Study quality was evaluated using the Newcastle-Ottawa Scale.
√	Assessment of heterogeneity	We used the I^2 value to assess heterogeneity.
√	Description of statistical methods in sufficient detail to be replicated	A meta-analysis was performed using Review Manager 5.3 software (Cochrane Collaboration). We calculated pooled risk ratio (RR) and 95% confidence intervals (95% CI) using a random effects model for dichotomous outcomes. Funnel plots were used to visually inspect for publication bias. Meta-regression was performed by the software Stata 14.0
√	Provision of appropriate tables and graphics	Search criteria for individual databases,flowchart for literature screening, data abstraction tables, funnel plots and forest plots for all analyses were all included.
Reporting of results should include		
√	Graph summarizing individual study estimates and overall estimate	Please refer to Figures 2, Figure 3 and Table 3
√	Table giving descriptive information for each study included	Please refer to Table 1.
√	Results of sensitivity testing	Please refer to Table 4, Appendix Table 3.
√	Indication of statistical uncertainty of findings	We considered an unadjusted, two-sided $p < 0.05$ to be statistically significant.
Reporting of discussion should include		
√	Quantitative assessment of bias	The quality of included studies were assessed by using the Newcastle Ottawa Scale (NOS)
√	Justification for exclusion	Exclusion criteria are detailed in the Methods section. Most studies were excluded because they were unrelated to the topic, or they did not directly compare frail and non-frail patient populations(refer to Appendix Table 1).
√	Assessment of quality of included studies	Please refer to Table 2
Reporting of conclusions should include		
√	Consideration of alternative explanations for observed results	We discuss that all included studies were observational and therefore at increased risk of bias. Furthermore, there was limited pooling of unadjusted data to account for cofounders.
√	Generalization of the conclusions	This systematic review and meta-analysis demonstrated frailty was significantly associated with the increased risk of clinical adverse events (all- cause mortality, COVID-19 severity, admission to intensive care unit, application of invasive mechanical ventilation, long-length stay)
√	Guidelines for future research	The result in our studies is needed to be validated in further well-designed perspective cohort studies. However, given the epidemic of COVID-19 and shortage of medical resources, paying more attention to screening frailty would contribute to disease management and resource allocation among patients with COVID-19.the result is needed to be validated in further well-designed perspective cohort studies
√	Disclosure of funding source	There was no funding source

2. Search strategy.

MEDLINE (inception – October 2020).

('frailty':ab,ti OR frail*:ab,ti) AND ('covid-19':ab,ti OR 'sars-cov-2':ab,ti OR '2019 novel coronavirus disease':ab,ti OR 'covid19':ab,ti OR 'covid-19 pandemic':ab,ti OR 'sars-cov-2 infection':ab,ti OR 'covid-19 virus disease':ab,ti OR '2019 novel coronavirus infection':ab,ti OR '2019-ncov infection':ab,ti OR 'coronavirus disease 2019':ab,ti OR 'coronavirus disease-19':ab,ti OR '2019-ncov disease':ab,ti OR 'covid-19 virus infection')

Pubmed (inception – October 2020).

((COVID-19[MeSH Terms]) OR (SARS-CoV-2[Title/Abstract]) OR (2019 novel coronavirus disease[Title/Abstract]) OR (COVID19[Title/Abstract]) OR (COVID-19 pandemic[Title/Abstract]) OR (SARS-CoV-2 infection[Title/Abstract]) OR (COVID-19 virus disease[Title/Abstract]) OR (2019 novel coronavirus infection[Title/Abstract]) OR (2019-nCoV infection[Title/Abstract]) OR (coronavirus disease 2019[Title/Abstract]) OR (coronavirus disease-19[Title/Abstract]) OR (2019-nCoV disease[Title/Abstract]) OR (COVID-19 virus infection[Title/Abstract])) AND ((frailty[MeSH Terms]) OR (frail*[Title/Abstract]))

Embase (inception – October 2020).

('frailty'/de OR frail*:ab,ti) AND ('covid-19':ab,ti OR 'sars-cov-2':ab,ti OR '2019 novel coronavirus disease':ab,ti OR 'covid19':ab,ti OR 'covid-19 pandemic':ab,ti OR 'sars-cov-2 infection':ab,ti OR 'covid-19 virus disease':ab,ti OR '2019 novel coronavirus infection':ab,ti OR '2019-ncov infection':ab,ti OR 'coronavirus disease 2019':ab,ti OR 'coronavirus disease-19':ab,ti OR '2019-ncov disease':ab,ti OR 'covid-19 virus infection')

3. Methods for calculating the odds ratio and its variance.

We consider a primary study *i* in which the association between smoking and invasive fungal disease (IFD) is estimated, diseased individuals being noted *D* and non–diseased individuals being noted *ND*. We can build the corresponding contingency table:

	Invasive fungal disease		
Smoking	Yes	No	*Sum*
Yes	a_i	b_i	a_{i+} b_i
No	c_i	d_i	c_{i+} d_i
Sum	$a_i + c_i = D_i$	$b_i + d_i = ND_i$	$a_i + c_i + b_i + d_i$

We consider the odds ratio (OR) as the outcome of interest.

By definition, the OR can be expressed as:

$$OR_i = \frac{a_i d_i}{b_i c_i}$$

The distribution of the natural logarithm of the OR is approximately normal and the standard error for *log(OR)* is approximately:

$$SE\left(log(OR_i)\right) = \sqrt{\frac{1}{a_i} + \frac{1}{b_i} + \frac{1}{c_i} + \frac{1}{d_i}}$$

4. Supplemental Tables

Table 1. Studies excluded after full-text evaluation.

Study	Reasons for exclusion
Saccoa M 2020[1]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Recinella G 2020[2]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Fasano A 2020[3]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Al Rihani SB 2020[4]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Knights H 2020[5]	the definition of frailty was unclear
Thakrar A 2020[6]	the data about the prevalence and outcomes of frailty among patients with COVID-19 could not be extracted
Brill SE 2020[7]	the data about the prevalence and outcomes of frailty among patients with COVID-19 could not be extracted
Doglietto F 2020[8]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Orlandi M 2020[9]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Ekiz T 2020[10]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Graham N 2020[11]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
OVERSIKT K 2020[12]	the language of paper was not English or Chinese
HOEK R 2020[13]	the data about the prevalence and outcomes of frailty among patients with COVID-19 could not be extracted
Landi F 2020[14]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Tay HS 2020[15]	Study design was not eligible: case report
Andrea SD 2020[16]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Porcheddu R 2020[17]	The study did not investigate the role of frailty in the context of COVID-19 epidemic
Crespo M 2020[18]	The definition of frailty was unclear
Cobos-Siles, M.[19]	The study did not investigate the relationship between frailty and the outcomes of COVID-19
Woolford SJ[20]	The study did not investigate the relationship between frailty and the outcomes of COVID-19
Zazzara, M. B.[21]	Cross-sectional study
Kokoszka-Bargieł, I.[22]	Cross-sectional study
Del Brutto, O. H.[23]	Cross-sectional study
Collins, J. T.[24]	Cross-sectional study
Palermo S 2020[25]	Review
Flaatten H 2020[26]	Review
Bossù P 2020[27]	Review
Bektas A 2020[28]	Review
Landi F 2020[29]	Review
Drury J 2020[30]	Review
Andrew MK 2020[31]	Review
Lidoriki I 2020[32]	Review
Leo DD 2020[33]	Review
Moccia F 2020[34]	Review
Michels G 2020[35]	Review
Hubbard RE 2020[36]	Review
Nickel Christian H 2020[37]	Review
Silver JK 2020[38]	Review
Roland K 2020[39]	Review
Groeger JL 2020[40]	Review
Deng MC 2020[41]	Review

Abbatecola AM 2020[42]	Review
Okamura T 2020[43]	Letter
Labenz C 2020[44]	Letter
Recinella G 2020[45]	Letter
Kenneth J 2020[46]	Letter
Bellelli G 2020[47]	Letter
Isaia G 2020[48]	Letter
Edward C 2020[49]	Letter
Boccardi V 2020[50]	Letter
Sinclair AJ 2020[51]	Letter
Kayani B 2020[52]	Letter
McCarthy K 2020[53]	Letter
Bruno G 2020[54]	Letter
Cheong CY 2020[55]	Letter
Vincent JL 2020[56]	Letter
Ma L 2020[[57]	Letter

Table 2. The definition of frailty in included studies.

Study	Frailty definition for the study
Chinnadurai R 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Kundi H 2020	The frailty was assessed by using the "Hospital Frailty Risk Score" (HFRS).The calculation of HFRS was based on one or more of 109 ICD-10-CM diagnosis codes recalled from all diagnosis codes of any hospitalization within the last two years. According to calculated HFRS, patients were classifified into three frailty risk groups based on previously validated cut points as low (<5 points), intermediate (5-15 points), and high (>15 points). Additionally, patients who had the HFRS of ≥5 were defifined as frail.
Steinmeyer Z 2020	Frailty was assessed by Frail Non-Disabled questionnaire.Patients were defined accordingly as robust, frail, or dependent.
Shi SH 2020	A frailty index (range 0–1, 1 indicates more severe frailty) was calculated using 35 items from the MDS, using a standard deficit accumulation approach. Variables used in construction of the frailty index include comorbidities, functional activities of daily living, sensory impairment, weight loss, and cognition.The clinical characteristic results were stratifified by frailty status, categorized by calculated frailty index11: Pre-frail (<0.25), mildly frail (0.25–0.34), or moderate-severely frail (≥0.35).
Hägg S 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Maguire D 2020	Frailty status was determined using the clinical frailty scale (CFS).
Ma Y 2020	Frailty level was identified by the number of criteria met. Individuals with none were considered "non-frail"; those meeting one or two criteria were considered "pre-frail", and those with three to five criteria were defined as "frail".
Marengoni A 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Darren A 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Miles A 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Owen RK 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Fiorentino M 2020	The Palliative Performance Scale is a validated tool to assess frailty. Scores range from 0 to 100. Prior studies in seriously ill individuals have used the Palliative Performance Scale as a measure of frailty. In these studies, a score ⩽70 was predictive of inhospital mortality and poor functional outcome at discharge, therefore, we dichotomized Palliative Performance Scale scores as low (⩽70) and high (>70)
Smet RD 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Burns GP 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Hewitt J 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.
Rawle MJ 2020	Frailty status was determined using the clinical frailty scale (CFS).Any patient with a score of five and above on the CFS was defined as being frail.

Table 3 Assessment of study quality using Newcastle-Ottawa quality assessment scale for cohort studies

Study	Selection (1)				Comparability (2)	Outcome (3)			Score
	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure to implants	Definition of controls	Demonstration that outcome of interest was not present at start of study	Assessment of outcome	Was follow up long enough for outcomes to occur	Adequacy of follow up of cohorts	
Chinnadurai R 2020	★	★	★	★	★★	★		★	8
Kundi H 2020	★	★	★	★	★★	★		★	8
Steinmeyer Z 2020	★	★	★	★		★		★	6
Shi SH 2020	★	★	★	★		★		★	6
Hägg S 2020	★	★	★	★	★	★		★	7
Maguire D 2020	★	★	★	★		★	★	★	7
Ma Y 2020	★	★	★	★	★★	★	★	★	9
Marengoni A 2020	★	★	★	★		★			5
Darren A 2020	★	★	★	★	★★	★		★	8
Miles A 2020	★	★	★	★	★★	★	★		8
Owen RK 2020			★	★	★★	★	★	★	7
Fiorentino M 2020	★	★	★		★★	★	★		7
Smet RD 2020	★	★	★	★	★★	★		★	8
Burns GP 2020	★	★	★		★★	★	★	★	7
Hewitt J 2020	★	★	★	★	★★	★	★	★	9
Rawle MJ 2020		★	★	★	★★	★	★	★	8

Table 4. pooled analysis of association of frailty with other clinical outcomes(COVID-19 severity, admission to intensive care unit, application of invasive mechanical ventilation, long-length stay) among COVID-19.

Clinical events	RR	95% CI	I^2 (%)	P	Model	Egger test	Begger test
COVID-19 severity	7.47	(1.73,32.40)	-	<0.001	Fixed model	-	-
Admission to ICU	1.80	(1.19,2.71)	96.2	0.005	Random model	-	0.317
Application of invasive mechanical ventilation	1.57	(1.21,2.03)		0.001	Random model	-	0.317
long-length stay(>10day)	1.22	(1.07,1.40)	73.6	0.003	Random model	-	0.317

Abbreviation: RR, risk ratio; CI, confidence interval; ICU, intensive care unit

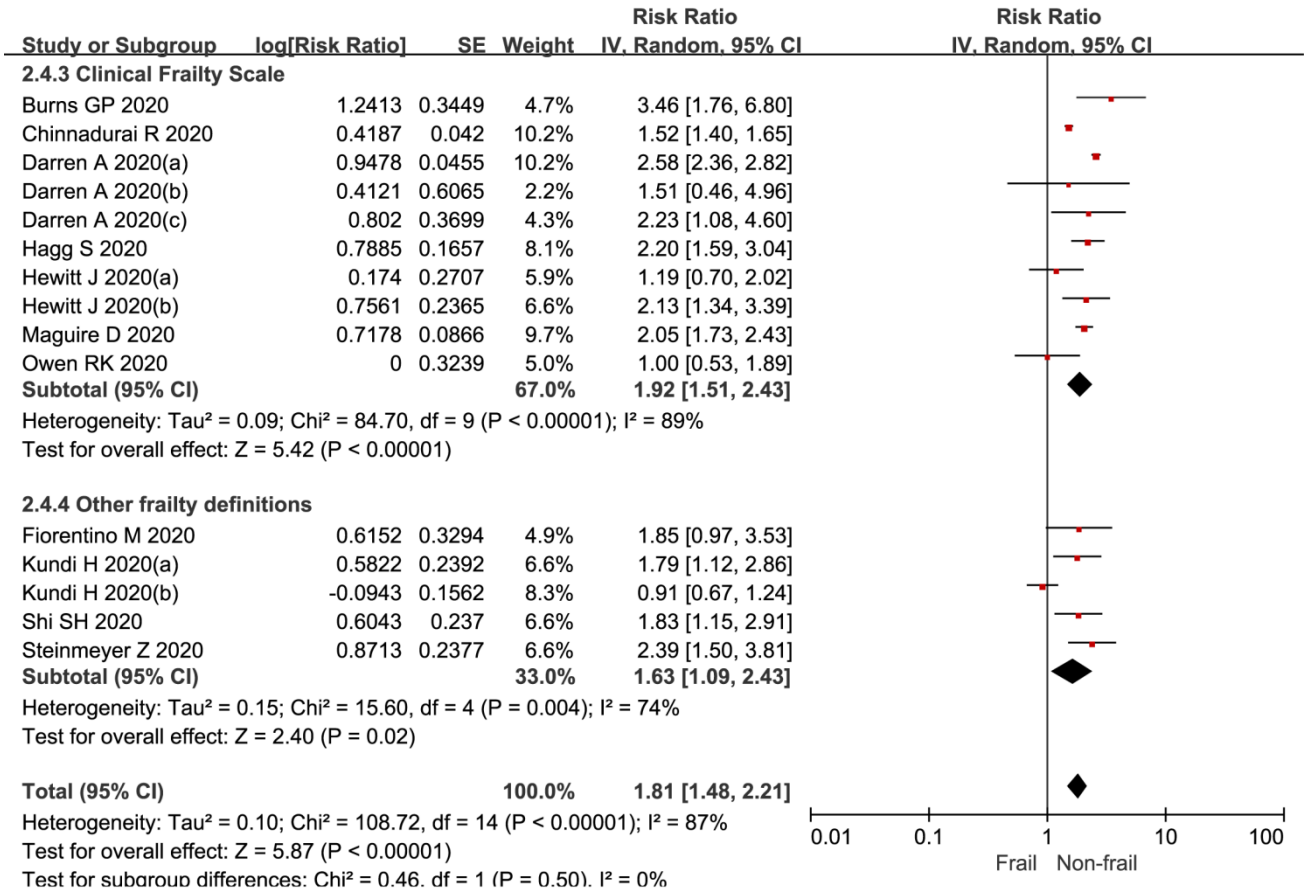
Table5. Sensitivity analysis: excluding included studies one by one.

Removed study	RR	95% CI	I^2 (%)	P	Model	Egger test	Begger test
Chinnadurai R 202	1.75	(1.43,2.15)	88.0	<0.001	Random model	0.607	0.511
Kundi H 2020	1.94	(1.59, 2.37)	86.0	<0.001	Random model	0.899	0.428
Steinmeyer Z 2020	1.77	(1.44, 2.18)	88.0	<0.001	Random model	0.768	1
Shi SH 2020	1.81	(1.47,2.22)	88.0	<0.001	Random model	0.705	0.827
Hagg S 2020	1.78	(1.44, 2.22)	88.0	<0.001	Random model	0.745	0.743
Maguire D 2020	1.79	(1.43, 2.23)	88.0	<0.001	Random model	0.689	1
Darren A 2020	1.71	(1.43,2.05)	72.0	<0.001	Random model	0.848	0.837
Owen RK 2020	1.87	(1.53,2.28)	88.0	<0.001	Random model	0.983	0.443
Fiorentino M 2020	1.81	(1.47, 2.22)	88.0	<0.001	Random model	1	0.724
Burns GP 2020	1.75	(1.43,2.15)	88.0	<0.001	Random model	0.885	0.913
Hewitt J 2020	1.77	(1.42,2.21)	89.0	<0.001	Random model	0.690	0.951

Abbreviation: RR, risk ratio; CI,confidence interval.

Supplemental Figures

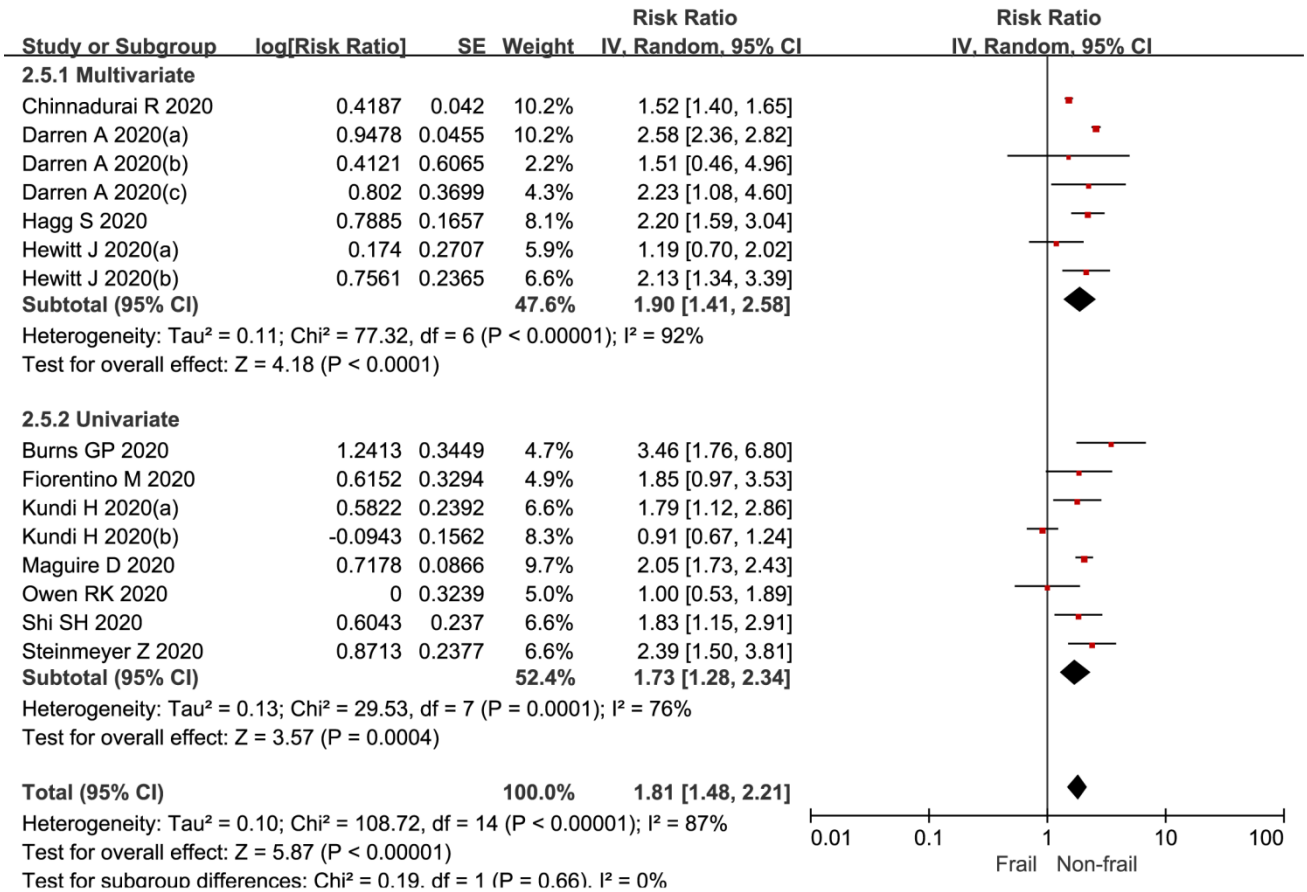
Figure 1- Stratified analysis by frailty definition



Abbreviation: RR, risk ratio.

Squares indicate risk ratios (RR) from primary studies, with sizes reflecting the statistical weight of the study in random-effects meta-analysis. The horizontal lines indicate 95% confidence intervals. Diamonds represent the summary RR and its 95% confidence interval. The vertical dashed line is aligned with the summary RR. The vertical black line shows the line of no effect (RR = 1). I-squared is the variation in RR attributable to heterogeneity rather than chance.

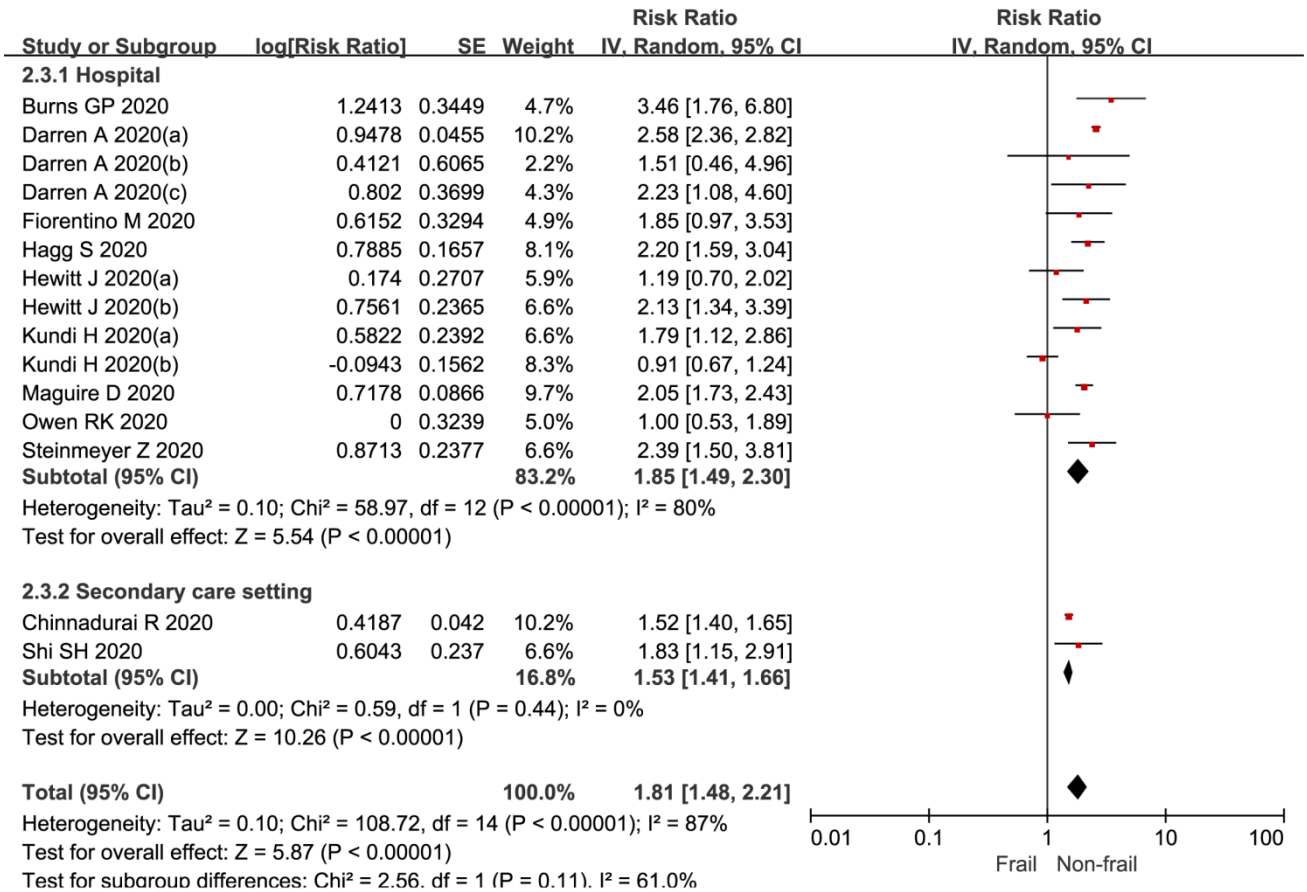
Figure 2- Stratified analysis by adjustment method



Abbreviation: RR, risk ratio.

Squares indicate risk ratios (RR) from primary studies, with sizes reflecting the statistical weight of the study in random-effects meta-analysis. The horizontal lines indicate 95% confidence intervals. Diamonds represent the summary RR and its 95% confidence interval. The vertical dashed line is aligned with the summary RR. The vertical black line shows the line of no effect (RR = 1). I-squared is the variation in RR attributable to heterogeneity rather than chance.

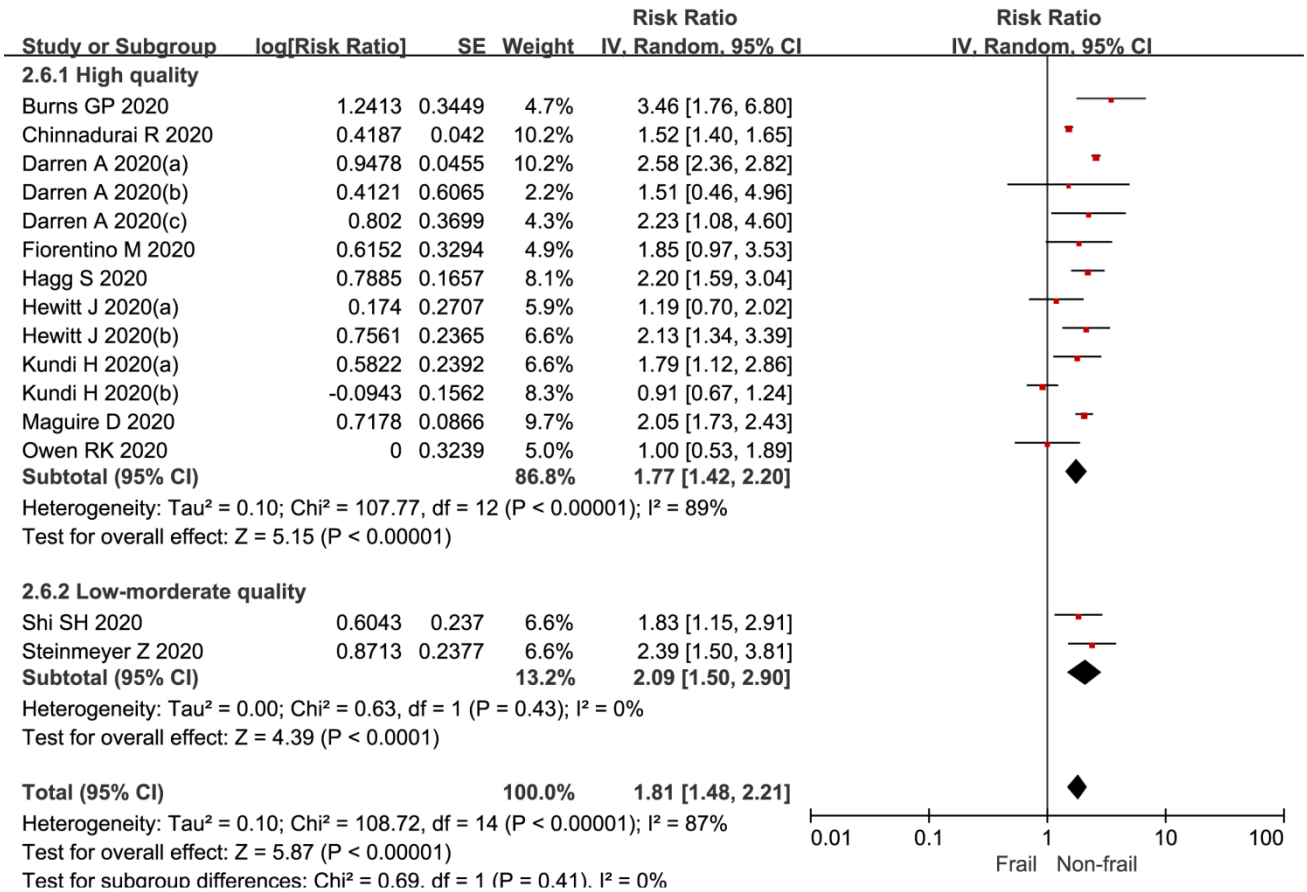
Figure 3- Stratified analysis by patients source



Abbreviation: RR, risk ratio.

Squares indicate risk ratios (RR) from primary studies, with sizes reflecting the statistical weight of the study in random-effects meta-analysis. The horizontal lines indicate 95% confidence intervals. Diamonds represent the summary RR and its 95% confidence interval. The vertical dashed line is aligned with the summary RR. The vertical black line shows the line of no effect (RR = 1). I-squared is the variation in RR attributable to heterogeneity rather than chance.

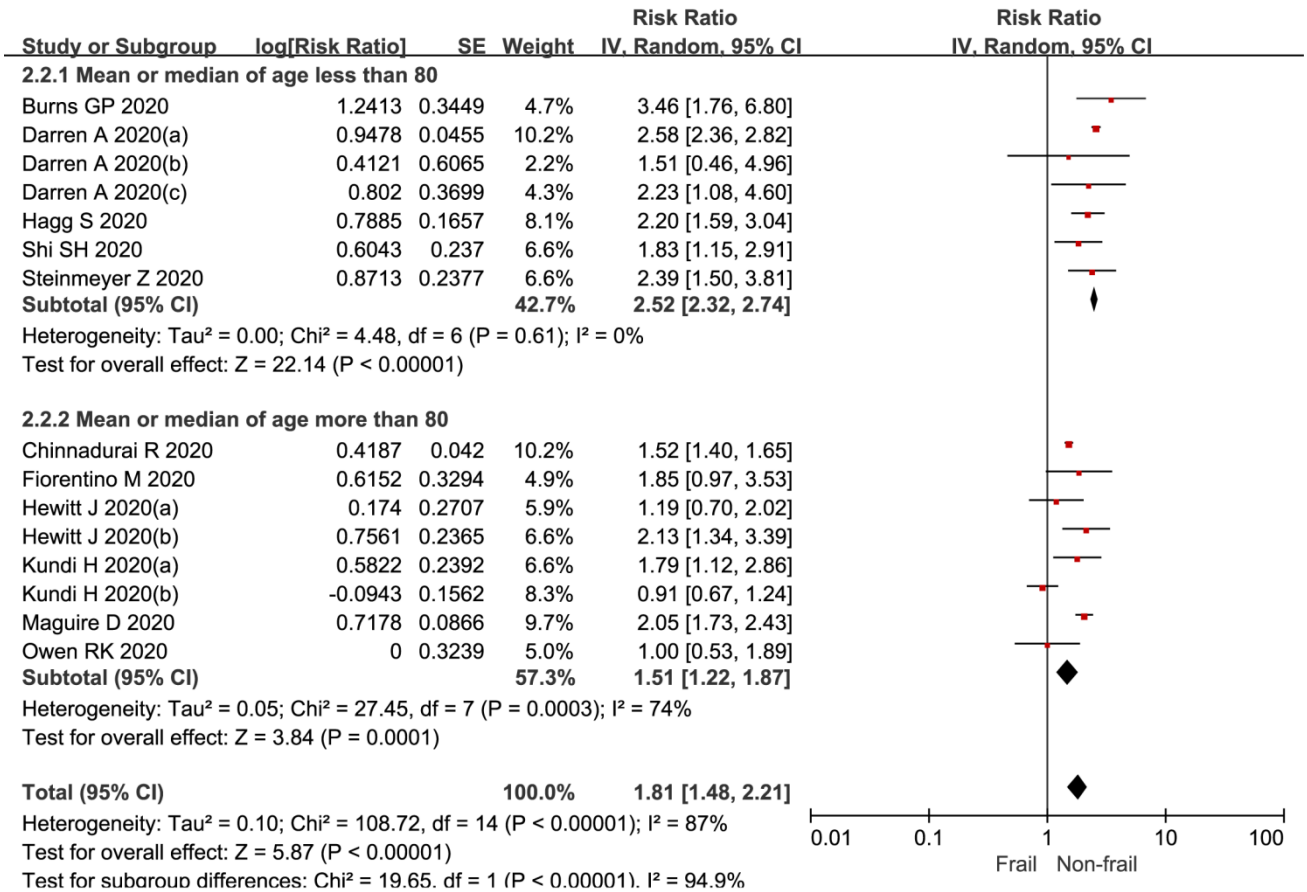
Figure 4- Stratified analysis by study quality



Abbreviation: RR, risk ratio.

Squares indicate risk ratios (RR) from primary studies, with sizes reflecting the statistical weight of the study in random-effects meta-analysis. The horizontal lines indicate 95% confidence intervals. Diamonds represent the summary RR and its 95% confidence interval. The vertical dashed line is aligned with the summary RR. The vertical black line shows the line of no effect (RR = 1). I-squared is the variation in RR attributable to heterogeneity rather than chance.

Figure 5- Stratified analysis by age



Abbreviation: RR, risk ratio.

Squares indicate risk ratios (RR) from primary studies, with sizes reflecting the statistical weight of the study in random-effects meta-analysis. The horizontal lines indicate 95% confidence intervals. Diamonds represent the summary RR and its 95% confidence interval. The vertical dashed line is aligned with the summary RR. The vertical black line shows the line of no effect (RR = 1). I-squared is the variation in RR attributable to heterogeneity rather than chance.

5. Additional references

1. Sacco G, Foucault G, Briere O, Annweiler C (2020) Findings and lessons from mass screening in a nursing home. Maturitas 141:46-52.
2. Recinella, Marasco G, Serafini G, Maestri L, Bianchi G, Forti P, Zoli M (2020) COVID-19: a monocentric study. Aging Clin Exp Res 1-7.
3. Fasano A, Elia AE, Dallocchio C, Canesi M, Alimonti D, Sorbera C,Alonso-Canovas A,Pezzoli G(2020) Predictors of COVID-19 outcome in Parkinson's disease. Parkinsonism Relat Disord 78: 134-137.
4. Al Rihani SB, Smith MK, Bikmetov R, Deodhar M, Dow P, Turgeon J, Michaud V.(2020) Risk of Adverse Drug Events Following the Virtual Addition of COVID-19 Repurposed Drugs to Drug Regimens of Frail Older Adults with Polypharmacy. J Clin Med 9:8
5. Knights H, Mayor N, Millar K, Cox M, Bunova E, Hughes M, Baker J, Mathew S, Russell-Jones D, Kotwica A (2020) Characteristics and outcomes of patients with COVID-19 at a district general hospital in Surrey, UK. Clin Med (Lond) 20(5): e148-e153.
6. Thakrar A, Chui K, Kapoor A, Hambidge J (2020) 30-day Mortality Rate of Patients with Hip Fractures during the COVID-19 Pandemic: A Single Centre Prospective Study in the United Kingdom. J Orthop Trauma 68:1199-1202
7. Brill SE,Jarvis HC,Ozcan E,Burns TLP, Warraich RA, Amani LJ, Jaffer A, Paget S,Sivaramakrishnan A,Creer DD (2020) COVID-19: a retrospective cohort study with focus on the over-80s and hospital-onset disease. BMC Med 18(1):194.
8. Doglietto F, Vezzoli M, Gheza F et al (2020) Factors Associated With Surgical Mortality and Complications Among Patients With and Without Coronavirus Disease 2019 (COVID-19) in Italy. JAMA Surg 155(8):1-14
9. Orlandi M, Lepri G,Bruni C et al (2020) The systemic sclerosis patient in the COVID-19 era: the challenging crossroad between immunosuppression, differential diagnosis and long-term psychological distress. Clin Rheumatol 39(7):2043-2047.
10. Ekiz T, Pazarlı AC (2020) Relationship between COVID-19 and obesity. Diabetes Metab Syndr 14(5): 761-763
11. Graham NSN,Junghans C, Downes R et al (2020) SARS-CoV-2 infection, clinical features and outcome of COVID-19 in United Kingdom nursing homes. J Infect 81(3):411-419.
12. Eriksen S, Grov EK, Lichtwarck B et al (2020) Palliative treatment and care for dying nursing home patients with COVID-19. Tidsskr Nor Laegeforen 140(8).
13. Hoek RAS, Manintveld OC, Betjes MGH et al (2020) COVID-19 in solid organ transplant recipients: a single-center experience. Transpl Int
14. Landi F,Barillaro C, Bellieni A et al (2020) The New Challenge of Geriatrics: Saving Frail Older People from the SARS-COV-2 Pandemic Infection. J Nutr Health Aging 24(5): 466-470
15. Tay HS,Harwood R (2020) Atypical presentation of COVID-19 in a frail older person. Age Ageing 49(4)523-524
16. D' Andrea S,Berardicurti O,Berardicurti A et al (2020) Clinical features and prognosis of COVID-19 in people with spinal cord injury: a case–control study. Spinal Cord Series and Cases 6(1)
17. Porcheddu R, Serra C,Kelvin D,Kelvin N,Rubino S (2020) Similarity in Case Fatality Rates (CFR) of COVID-19/SARS-COV-2 in Italy and China. Journal of Infection in Developing Countries 14(2):125-128
18. Crespo M, Pérez-Sáez MJ, Redondo-Pachón D et al (2020) COVID-19 in elderly kidney transplant recipients. Am J Transplant, 2020. 20(10):2883-2889.
19. Cobos-Siles M, Cubero-Morais P, Arroyo-Jiménez I et al (2020) Cause-specific death in hospitalized individuals infected with SARS-CoV-2: more than just acute respiratory failure or thromboembolic events. Intern Emerg Med, 2020: 1-12.
20. Woolford1 SJ, Angelo SD, Curtis1EM et al (2020) COVID-19 and associations with frailty and multimorbidity: a prospective analysis of UK Biobank participants. Aging Clinical and Experimental Research 32:1897–1905
21. Zazzara MB, Penfold RS, Roberts AL et al (2020) Probable delirium is a presenting symptom of COVID-19 in frail, older adults: a cohort study of 322 hospitalised and 535 community-based older adults. Age Ageing
22. Kokoszka-Bargieł I, Cyprys P, Rutkowska K et al (2020) Intensive Care Unit Admissions During the First 3 Months of the COVID-19 Pandemic in Poland: A Single-Center, Cross-Sectional Study. Med Sci Monit 26:e926974.
23. Del Brutto OH, Costa AF, Recalde BY et al (2020) Frailty and SARS-CoV-2 infection. A population-based study in a highly endemic village. J Neurol Sci 418:117136.
24. Collins JT,Short R,Carter B et al (2020) The Clinical Frailty Scale: Estimating the Prevalence of Frailty in Older Patients Hospitalised with COVID-19. The COPE Study. Geriatrics (Basel)5(3)
25. Palermo S (2020) Covid-19 Pandemic: Maximizing Future Vaccination Treatments Considering Aging and Frailty. Front Med (Lausanne) 7: 558835
26. Flaatten H, Beil M, Guidet B (2020) Guidet, Prognostication in older ICU patients: mission impossible? Br J Anaesth 125(5): 655-657
27. Bossù P, Toppi E, Sterbini V, Spalletta G (2020) Implication of Aging Related Chronic Neuroinflammation on COVID-19 Pandemic. J Pers Med 10(3)

28. Bektas A, Schurman SH, Franceschi C,Ferrucci L (2020) A public health perspective of aging: do hyper-inflammatory syndromes such as COVID-19, SARS, ARDS, cytokine storm syndrome, and post-ICU syndrome accelerate short- and long-term inflammaging? Immun Ageing 17: 23
29. Landi F,Barillaro C,Bellieni A et al (2020) The Geriatrician: The Frontline Specialist in the Treatment of COVID-19 Patients. J Am Med Dir Assoc 21(7):937-938.
30. Drury J,Reicher S,Stott C (2020) COVID-19 in context: Why do people die in emergencies? It's probably not because of collective psychology. Br J Soc Psychol 59(3): 686-693
31.Andrew M, Searle SD,McElhaney JE,McNeil SA,Clarke B,Rockwood K,Kelvin DJ (2020) COVID-19, frailty and long-term care: Implications for policy and practice. J Infect Dev Ctries 14(5): 428-432
32.Lidoriki I,Frountzas M,Schizas D (2020) Could nutritional and functional status serve as prognostic factors for COVID-19 in the elderly? Med Hypotheses 144:109946.
33. Leo DD, Trabucchi M (2020) COVID-19 and the Fears of Italian Senior Citizens. Int J Environ Res Public Health 17(10)
34. Moccia F, Gerbino A,Lionetti V et al (2020) COVID-19-associated cardiovascular morbidity in older adults: a position paper from the Italian Society of Cardiovascular Researches. Geroscience 42(4): 1021-1049
35. Michels G, Heppner HJ (2020) Heppner, [Prioritization of geriatric patients in care homes and residential homes in the context of the COVID pandemic]. MMW Fortschr Med 162(9):48-51
36. Hubbard RE,Maier AB,Hilmer SN,Naganathan V,Etherton-Beer C,Rockwood K (2020) Frailty in the face of COVID-19. Age Ageing 49(4):499-500
37.Nickel CH, Rueegg M,Pargger H,Bingisser R (2020) Age, comorbidity, frailty status: effects on disposition and resource allocation during the COVID-19 pandemic. Swiss Med Wkly 150: 20269
38. Silver JK (2020) Prehabilitation May Help Mitigate an Increase in COVID-19 Peripandemic Surgical Morbidity and Mortality. Am J Phys Med Rehabil 99(6): 459-463
39. Kunz R,Minder M (2020) Minder, COVID-19 pandemic: palliative care for elderly and frail patients at home and in residential and nursing homes. Swiss Med Wkly 150:w20235
40.Groeger JL,Starrels JL,Ehrlich AR (2020) Starrels, and A.R. Ehrlich, Older Adults with COVID-19 Can Choose Care at Home: Lessons Learned from New York City. Journal of General Internal Medicine 35(9): 2755-2756
41.Deng MC (2020) Multi-dimensional COVID-19 short- and long-term outcome prediction algorithm. Expert Review of Precision Medicine and Drug Development 5(4):239-242
42.Abbatecola AM,Antonelli-Incalzi R (2020) COVID-19 Spiraling of Frailty in Older Italian Patients. Journal of Nutrition.Health and Aging 24(5):453-455.
43.Okamura T,Ura C,Sugiyama M et al (2020) Defending community living for frail older people during the COVID-19 pandemic. Psychogeriatrics
44.Labenz C,Schattenberg JM,Wörns MA,Sprinzl MF (2020) Response: Frailty assessment in the COVID-19 pandemic. J Investig Med 68(7):1302
45. Recinella G,Marasco G,Di Battista A,Bianchi G,Zoli M (2020) Prognostic role of nutritional status in elderly patients hospitalized for COVID-19. Med Hypotheses144:110016
46. Ng Cheong Chung KJ,Kunadian V (2020) Frailty assessment in the COVID-19 pandemic. J Investig Med 68(7): 1300-1301
47.Bellelli G,Rebora P,Citerio G (2020) The role of frailty in COVID-19 patients. Intensive Care Med 46(10):1958-1959.
48. Isaia G, Marinello R,Tibaldi V,Tamone C, Bo M (2020) Atypical Presentation of Covid-19 in an Older Adult With Severe Alzheimer Disease. Am J Geriatr Psychiatry 28(7): 790-791
49. Chong E, Chan M,Tan HN, Lim WS (2020) COVID-19: Use of the Clinical Frailty Scale for Critical Care Decisions. J Am Geriatr Soc 68(6): E30-e32
50. Boccardi V,Ruggiero C,Mecocci P (2020) COVID-19: A Geriatric Emergency. Geriatrics (Basel)5(2)
51. Sinclair AJ,Abdelhafiz AH (2020) Age, frailty and diabetes - triple jeopardy for vulnerability to COVID-19 infection. EClinicalMedicine, 2020. 22:100343
52. Kayani B, Onochie E, Patil V (2020) The effects of COVID-19 on perioperative morbidity and mortality in patients with hip fractures. Bone and Joint Journal 102(10): 1279-1280
53. McCarthy K,Myint PK,Moug S et al (2020) Resumption of elective colorectal surgery during COVID-19 and risk of death. Colorectal Disease 22(9):1026-1027
54. Bruno G, Perelli S,Fabrizio C,Buccoliero GB (2020) Short-term outcomes in individuals aged 75 or older with severe coronavirus disease (COVID-19): First observations from an infectious diseases unit in Southern Italy. Journal of Infection 81(2): e86-e88.
55. Cheong CY,Yap PLK (2020) Staying True to the Calling of Geriatric Medicine Amid the Waves of COVID-19. Journal of Nutrition, Health and Aging 24(5):534-535
56. Vincent JL,Taccone FS (2020) Taccone, Understanding pathways to death in patients with COVID-19. The Lancet Respiratory Medicine 8(5):430-432

57. Ma L (2020) COVID-19 Prevention: Use of Self-Reported Tools to Screen Frail Older Adults. Journal of Nutrition, Health and Aging